

Demonstration of TactStyle: Generating Tactile Textures with Generative AI for Digital Fabrication

Faraz Faruqi
MIT CSAIL
Cambridge, MA, USA
ffaruqi@mit.edu

Yunyi Zhu
MIT CSAIL
Cambridge, MA, USA
yunyizhu@mit.edu

Maxine Perroni-Scharf
MIT CSAIL
Cambridge, MA, USA
max1@mit.edu

Shuyue Feng
Zhejiang University
Hangzhou, China
shuyuefeng@zju.edu.cn

Jaskaran Singh Walia
Vellore Institute of Technology
Chennai, India
karanwalia2k3@gmail.com

Donald Degraen*
HIT Lab NZ, University of Canterbury
Christchurch, New Zealand
donald.degraen@canterbury.ac.nz

Stefanie Mueller
MIT CSAIL
Cambridge, MA, USA
stefanie.mueller@mit.edu

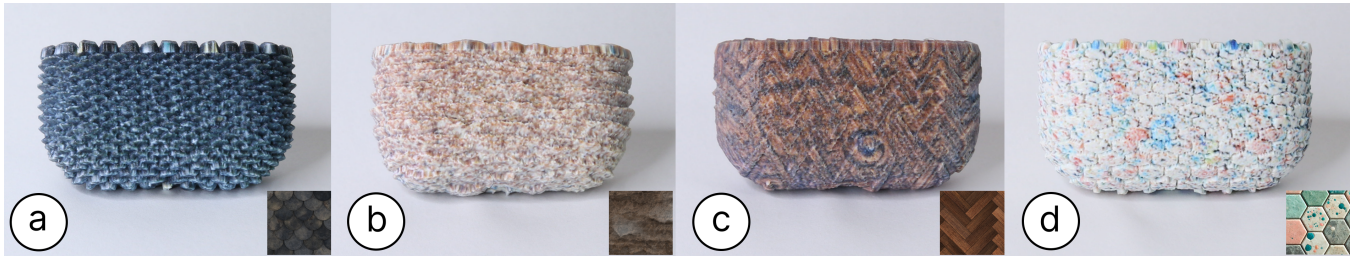


Figure 1: TactStyle allows creators to stylize 3D models with image input while incorporating the tactile properties of the texture in addition to its color. Here, we show four different textures applied to the same 3D model, an AirPods cover, with the image stylization prompt shown on the right bottom. The different textures used are: a) round stone roof, b) layered brown rock, c) herringbone wood, and d) colorful hexagonal tiles.

Abstract

Recent work in Generative AI enables the stylization of 3D models based on image prompts. However, these methods do not incorporate tactile information, leading to designs that lack the expected tactile properties. We present TactStyle, a system that allows creators to stylize 3D models with images while incorporating the expected tactile properties. TactStyle accomplishes this using a modified image-generation model fine-tuned to generate height-fields for given surface textures. By optimizing 3D model surfaces to embody a generated texture, TactStyle creates models that match the desired style and replicate the tactile experience. We present TactStyle’s user interface to create stylized 3D models with accurate textures, present multiple textured tiles for blind perception experience, and present five application scenarios with fabricated 3D models.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Personal Fabrication; Digital Fabrication; 3D Printing; Generative AI.

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1 Introduction

With the growing popularity of 3D printing research within HCI [17], there is also increasing interest in developing tools that enable users to customize 3D models. Open-source repositories, such as Thingiverse [4], are a useful resource for ready-to-print 3D models, however, their customization is limited to changing predefined parameters [1, 7]. Recent advances in Generative AI allow users to more freely customize their 3D models using text prompts or images as user-provided style descriptions [8, 16]. However, these existing frameworks for stylizing 3D models primarily

*Also with HCI Group, University of Duisburg-Essen.

focus on modifying the model to match a desired visual appearance as described via the provided text or image prompt [18].

One underexplored area of model customization is texture—specifically, the ‘tactile feedback’ of printed structures, such as whether a surface feels smooth, rough, or potentially reminiscent of materials like wood grain or stone. Our ability to sense these textures through touch plays a crucial role in our interactions with the physical world, shaping how we perceive and manipulate objects and influencing our emotional and cognitive responses to them [10]. Therefore, augmenting printed structures with appropriate tactile properties can enrich interaction with physical objects, especially when mimicking materials that differ from the 3D printing material.

In this demonstration, we present TactStyle, a system that allows creators to stylize 3D models with images while incorporating the expected tactile properties. TactStyle accomplishes this by separating the visual and the geometry stylization, and augmenting the process with a novel geometry stylization module that replicates the tactile properties of textures based on user input. The novel geometry stylization module uses a fine-tuned variational autoencoder (VAE) [12], that translates the user provided visual image of a texture into a surface microgeometry or heightfield. The model then uses this heightfield to manipulate the geometry to create the tactile properties on the 3D model surface. Separately, the visual appearance of the 3D model is optimized with a method [8] that has been shown to accurately replicate visual qualities. Thus, by optimizing 3D model surfaces to embody both the color and tactile properties of a given input image, TactStyle allows creators to generate stylized models that not only visually match the desired style but also replicate the tactile experience.

2 System Overview

Prior work [8, 9, 14] has shown that Generative-AI based stylization methods provide a close approximation of the user’s style. However, our formative study found that geometry modifications do not accurately replicate the desired texture represented by its surface microgeometry. To enable the replication of a surface’s microgeometry, and by extension, its tactile properties, we designed TactStyle to focus on the geometry modification part of the stylization process, using Style2Fab [8] as the color optimization method.

TactStyle augments existing stylization methods with a new approach to modifying the geometry of 3D models to replicate the tactile properties of the texture described by the user. It accomplishes this by co-optimizing the geometry and the color channels separately. We call these two modules: (1) the color stylization module and (2) the geometry stylization module. The focus of this paper is on the geometry stylization module.

Our main challenge was to design a geometry stylization module that modifies a 3D model’s geometry to replicate the tactile properties of a texture. We approach this problem as an image-generation problem, where the goal is to generate heightfields, given the image of a texture. TactStyle accomplishes this by fine-tuning an image generation model to generate heightfields based on visual images of a texture. This heightfield is then used to modify the 3D model’s geometry using an approach based on UV mapping. Thus, the color and geometry stylization modules work in tandem, stylizing the

color and geometry of the 3D model to replicate both the visual appearance and tactile properties of a given texture (Figure 2a).

3 Heightfield Generation Technique

In this section, we describe our novel heightfield generation model. This model takes a texture image as a prompt and generates the associated heightfield. For this purpose, we fine-tune a trained diffusion-based Image-to-Image model and integrate it into the TactStyle system. In the following subsections, we describe the modified architecture of the diffusion model and the dataset used to train and test.

3.1 Diffusion Model

We approach this problem as an image generation task and use a modified version of the Stable Diffusion model [19], a popular open-source image generation model. Specifically, we use an image-to-image generation model proposed in SDEdit [15]. This deep-learning model uses a diffusion model to synthesize new realistic images. Our goal with this model was to generate a heightfield given an image of a texture. Since heightfields are traditionally represented as grayscale images, we: (1) modify the VAE architecture to generate grayscale images, i.e., heightfields, (2) and fine-tune the trained model on our texture image-heightfield pairs.

3.2 Modified Model Architecture

We use a trained open-source Image-to-Image Generation model available through the Diffusers library [20]. This model’s essential component is the VAE, which encodes an image into a latent representation and then decodes it into another image. We modify the architecture’s decoder module by adding 4 additional layers to learn heightfield features and modify the final layer to output single-channel grayscale images.

The Variational Auto Encoder (VAE) is fine-tuned for generating accurate heightfields using the PBR Dataset [3] consisting of texture image-heightfield pairs. The associated heightfields serve as ground truth representations of the tactile features, and our model learns the correlation between visual appearance and tactile properties.

To apply the AI-generated textures to 3D models, we apply the heightmap by displacing vertices along their normals based on the corresponding height values from a UV map normalized to fit the texture map, producing a texturized object ready for 3D printing. This process creates a final texturized object that is ready to be 3D printed, as shown in geometry stylization step of Figure 2a.

4 User Interface and Workflow

TactStyle has been implemented as a plugin for the open-source 3D design software tool Blender [5] to allow easy integration with makers’ existing workflows. Figure 4 shows a view of the interface. To stylize a model with TactStyle, the user (1) loads their model, (2) uploads the image prompt of the desired texture, and (3) clicks the stylize button. TactStyle then processes the model and stylizes it using the integrated color and geometry simulation modules. The stylized model is rendered next to the original model, which the user can export for fabrication.

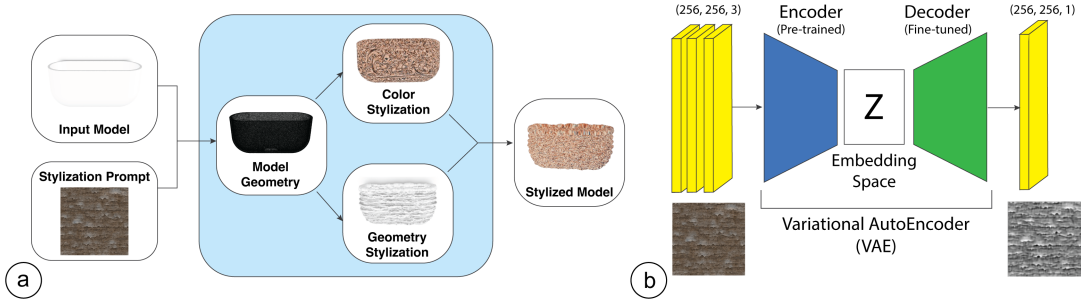


Figure 2: TactStyle augments traditional 3D model stylization techniques by introducing a novel geometry stylization module that replicates the tactile properties of textures based on user input. (a) The system takes an input model and a stylization prompt (e.g., an image of a texture) and applies two separate stylization processes: (1) Color Stylization and (2) Geometry Stylization. The color stylization modifies the model’s visual appearance, while the geometry stylization alters its surface to reflect tactile properties. The two modules operate in tandem, creating a stylized 3D model that replicates both the visual and tactile aspects of the texture. (b) The geometry stylization module uses a variational autoencoder (VAE) to generate heightfields from texture images, which are then applied to modify the model’s surface geometry, enabling co-optimization of geometry and color for a unified tactile and visual experience.

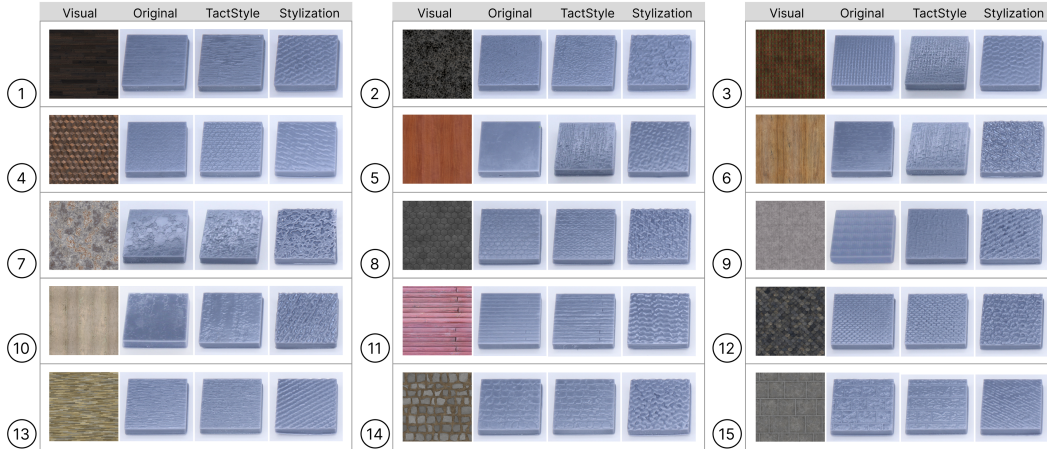


Figure 3: 3D-Printed samples of 15 textures to be presented during the demonstration. There are four sets of models: ‘visual set’, ‘original set’, ‘TactStyle set’, and the ‘stylization set’. The original set was created with the heightfield associated with the texture and served as the groundtruth. The reconstructed set was created using TactStyle, with the texture image as input. The visual set was created using printed texture images. Finally, the baseline set was created using Style2Fab, using the texture image as input.

4.0.1 Preprocessing. Once the user has loaded an OBJ file of their 3D mesh into the plugin, the model is automatically pre-processed for stylization.

4.0.2 Stylization. TactStyle used two modules — the color stylization module and the geometry stylization module to optimize both visual and tactile properties. As shown in Figure 2, TactStyle uses Style2Fab [8] for iterative color optimization where the model’s color channels are modified to approximate the style in the image. Next, the geometry stylization module uses the modified image generation model to generate a heightfield using the texture image prompt provided by the user. This heightfield is applied to the

model. The completed model is rendered alongside the original model for review.

4.0.3 Fine-Tuning and Export. Users can iterate on this process and apply new styles using new image prompts as needed. Users can also optionally increase the amount of height displacement to magnify their texture by changing the ‘Texture Magnification Factor’ slider shown in Figure 4e which can exaggerate or diminish the texture applied. This factor is by default at 1.0, which was used in our study following the height displacement values from Degraen et al. [6]. Finally, the user can export the stylized model and fabricate it.

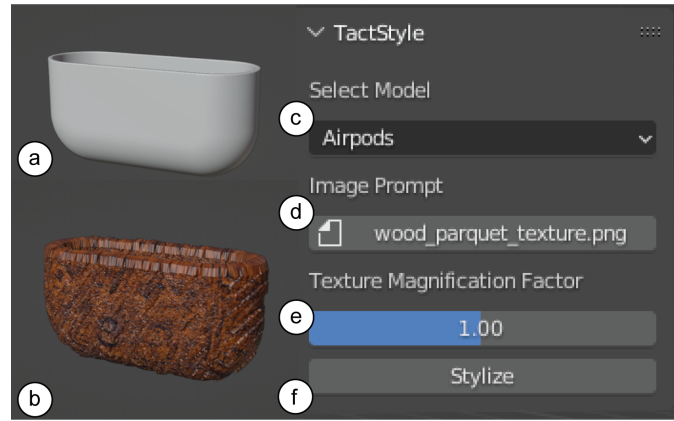


Figure 4: TactStyle’s user interface, implemented as a Blender plugin, allows users to load (a) original 3D models and (b) stylize with image prompts. In order to use TactStyle, the user (c) loads the model, (d) uploads an image of their desired texture, (e) optionally adjust the Texture Magnification Factor to control the level of height displacement applied on the 3D model. (f) Finally, the user clicks the “Style” button, which starts the stylization process using TactStyle’s integrated color and geometry stylization modules.



Figure 5: Application Examples for TactStyle: a) a phone stand stylized with a wood-parquet texture, b) a granite-textured vase, c) an airpods case stylized with a texture of ‘round stone roof’ and ‘layered brown rock’, d) two tiles, one styled with a volcanic rock texture and the other styled with stone from the Grand Canyon, e) walking stick handle stylized with a rough rock texture.

5 Applications

In this section, we showcase how TactStyle’s stylization technique allows users to stylize 3D models with accurate tactile properties for fabrication. We demonstrate five application scenarios across four categories: home decor, personal accessories, tactile learning tools, and personalized health applications. All 3D models shown in Figure 5 were stylized with TactStyle using textures not present in the training dataset and printed on a Stratasys J55 printer.

5.1 Home Decor

TactStyle can be used to apply textures to 3D models, enabling users to enhance the tactile experience of 3D-printed items. This allows individuals to create customized, textured versions of everyday objects, adding both aesthetic value and functional grip or surface texture. We illustrate several two applications of TactStyle

in applying textures to functional home objects. Figure 5a shows a *wood-parquet* textured phone stand, demonstrating how organic textures can be applied to enhance the visual appeal and usability of frequently handled items. Figure 5b, shows a granite-textured vase. By combining TactStyle and digital fabrication techniques, users can personalize their objects, or prototype specific tactile properties in addition to the aesthetics of their home decor objects.

5.2 Personalizing Accessories

Personal accessories are a popular domain for personalized fabrication. In Figure 5c we showcase an AirPods case, stylized with two different textures: a round stone roof texture taken from an image of a roof (top), and another stylized with an image from the Grand Canyon (bottom). These textures not only provide visual distinction but also have different surface microgeometry, associated with the texture.

5.3 Tactile Learning Tools

TactStyle has the potential to create educational tools that enhance learning in subjects such as geometry, topography (e.g., the texture of different terrains), biology (e.g., the texture of animal skins), and geography. To exemplify this concept, we present two examples in Figure 5d: the top surface features a ‘volcanic rock texture’, and the bottom surface replicates the texture of stone from ‘the Grand Canyon’. Tangible learning materials are well-known to improve educational outcomes, particularly by engaging multiple senses [11]. TactStyle offers a new way to create such materials, allowing educators to bring textures and surfaces to life in the classroom.

5.4 Customizable Assistive Devices

In the field of “Medical Making” [13] and “DIY Assistive Technology” [2] personalized fabrication by nontechnical experts is becoming an emerging and critical domain. TactStyle can be employed to customize assistive devices with specific textures, enhancing grip, comfort, or usability tailored to the unique needs of the user. Figure 5e illustrates this by applying a ‘rough rock’ texture to the handle of a walking stick. This texture generates a rough heightfield, which post-fabrication, significantly increases surface friction, thereby improving grip and stability for the user. Such tactile enhancements are particularly valuable for assistive devices, where safety and ease of use are critical.

6 Conclusion

In this demonstration, we present TactStyle, a system that allows users to stylize 3D models using image prompts, replicating both visual appearance and tactile properties. By extending generative AI techniques, TactStyle generates tactile features as a heightfield and applies them to 3D models. We present TactStyle’s user interface, demonstrate AI-generated textures on tiles for perception, and present five application scenarios with fabricated 3D models. Our findings show that TactStyle successfully aligns visual and tactile properties, enabling more realistic 3D model stylization. This work opens up new possibilities in cross-modal design, and future work can expand TactStyle by incorporating material descriptors to further enhance its tactile accuracy.

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